Singapore Institute of Technology
Information and Communications Technology Cluster

ICT4011 Capstone Project

<u>Capstone Project Details (Form B)</u>

A. General Capstone Project Information

Project Information:

(Project Title): Multi-agent GenAI for query large dataset to generate report and dashboard

(Organization): ST Engineering Urban Solutions Ltd.

(Initiation):

☒ Project initiated by industry
☐ Project initiated by student

(Multi-disciplinary):

☒ Project is multi-disciplinary (e.g. ICT and ENG)
☐ Project is single-disciplinary (e.g. ICT only)

Candidate Particulars:

(Name of Candidate): Tay Siang Long

(Matriculation Number): 2203190

(Programme):

☒ B.Eng. (Hons) in Information and Communications Technology (Information Security)
☐ B.Eng. (Hons) in Information and Communications Technology (Software Engineering)

(Project Period): *Project start date* : 5 May 2025 to *Project end date*
: 12 April 2026

Main Academic Supervisor Details:

(Name): Spiridon Bakiras

(Email Address): spiridon.bakiras@singaporetech.edu.sg

(Contact Number): *The academic supervisor's contact number*

(Designation): *The academic supervisor's designation*

Industry Supervisor Details:

(Name): Wu Yingjie

(Email Address): yingjie.wu@stengg.com

(Contact Number): 88517312

(Designation): *The industry supervisor's designation*

(Department / Division): *The industry supervisor's department / division*

B. Capstone Project Details

Project Overview:
(Please use a separate sheet if necessary)

The capstone project, 'Multi-Agent System for Query-to-Dashboard Pipeline,' aims to develop an intelligent system that automates the process of generating reports and dashboards from natural language queries. The system will use a multi-agent framework with a Model Context Protocol (MCP) server to provide tool context and analysis functions to language models.

A primary focus is on quantitative analysis, insight generation, report composition, and dashboard rendering. However, as development progresses, if the Analysis Agent proves capable of fully performing the functions of the Insight, Report Generation, and Dashboard Agents, these may be modularised under the Analysis Agent. This would result in a streamlined architecture with potentially only one or two final agents.

Project Objectives:
(Please use a separate sheet if necessary)

Develop a modular Analysis Agent that integrates four key capabilities: statistical and mathematical analysis, insight interpretation, report generation, and dashboard rendering. Each function will be implemented as an internal module, allowing focused development and testing while enabling future scalability.

Enable an automated pipeline that takes in natural language queries and produces structured reports and dashboards. This will involve minimal upstream agents for query understanding and SQL translation, feeding directly into the Analysis Agent for processing and output generation.

Build a multi-agent or modular mono-agent framework that supports flexible integration via the Model Context Protocol (MCP). This ensures reusable tools and consistent context handling across all modules or agents, whether deployed separately or consolidated.

Execute structured testing and iterative refinement across the entire system. Core components will be evaluated using defined datasets and query types, with failure cases documented and used to guide improvements.

Allow dynamic architectural optimization by merging agents where overlapping functionality is identified. The final architecture may consolidate multiple agents into the Analysis Agent if it proves capable of handling all required tasks independently.

Industry Relevancy:
(Please use a separate sheet if necessary)

This project holds significant relevance for ST Engineering and the transportation/smart city industry at large. Given ST Engineering's extensive work with traffic, car parking, and public transport data across Singapore and internationally.

Empowering Data-Driven Decisions: The system will enable ST Engineering's analysts and operational teams to quickly generate custom reports and dashboards from natural language queries, reducing reliance on specialized technical skills for data access. This significantly speeds up decision-making processes for traffic management, urban planning, and operational efficiency.

Automating Insights from Complex Datasets: By focusing on the Quantitative Analysis Agent, Insight Agent, Report Generation Agent, and Dashboard Generation Agent, the project directly addresses the challenge of extracting actionable intelligence from vast and complex transportation datasets. It automates the process of identifying trends, anomalies, and correlations in traffic flow, parking occupancy, or public transport ridership, turning raw data into strategic insights.

Improving Operational Efficiency: Automating the generation of performance metrics, incident reports, or usage patterns for transportation networks can free up significant manpower, allowing ST Engineering to reallocate resources to higher-value tasks and improve overall operational efficiency across its various smart mobility solutions.

Scalability and User Accessibility: The multi-agent architecture provides a scalable solution for handling diverse data analysis requests. Its natural language interface enhances accessibility for a wider range of users within the organization, promoting a more data-literate environment within ST Engineering's operations.

Project Scope:
(Please use a separate sheet if necessary)

This project focuses on building an intelligent multi-agent or modular system that automates query-driven data analysis, insight extraction, report generation, and dashboard rendering. Central to the system is the Analysis Agent, which will be the primary component responsible for processing and interpreting data. As the system evolves, this agent may substitute or absorb the roles of the Insight, Report Generation, and Dashboard agents, resulting in a more streamlined and maintainable architecture.

The Analysis Agent will perform key analytical functions such as trend detection, anomaly identification, comparative analysis, and data distribution evaluation. It will also interpret these findings to generate actionable insights, compile them into structured, query-relevant reports, and visualize the results through interactive dashboards.

To support modularity and reuse, the Model Context Protocol (MCP) will act as a centralized registry for reusable tools and functions. This ensures consistent execution, easy tool discovery, and standardized interfaces across agents. While upstream agents—such as those handling query understanding and SQL generation—will be included, they will be minimally implemented, serving only to support the core pipeline and enable full demonstration of the query-to-dashboard system.

Limitations of the project:

Novelty of Model Context Protocol (MCP): As MCP is a relatively new concept or framework, its implementation may involve navigating evolving best practices, limited readily available examples, and potential challenges in integration or optimization not typically encountered with more mature technologies.

Project Outputs and Deliverables
(Please use a separate sheet if necessary)

The primary deliverable will be a working prototype of a query-to-dashboard system capable of handling natural language queries and producing structured analytical outputs. The system will feature a fully functional Analysis Agent that performs statistical analysis, interprets results, generates reports, and renders dashboards. Depending on development progress, this agent may also consolidate the roles of the Insight, Report Generation, and Dashboard Agents into a single, modular component.

To support the full pipeline, simplified upstream agents will be implemented to handle basic query parsing and SQL generation. These will facilitate system-level integration and allow the end-to-end flow—from user input to dashboard output—to be demonstrated effectively.

The system will also include an MCP (Model Context Protocol) registry, which holds reusable analytical functions and tools. This registry will enable consistent, context-aware function invocation across all agents or modules.

Deliverables will include evaluation results, structured test cases, logs, and documentation of iterative improvements made during development. Ultimately, if agent consolidation proves feasible, the final architecture may consist of only one or two core agents, optimized for performance, maintainability, and scalability.

C. Knowledge and Training Requirements

Applicable Knowledge from the Degree Programme

(Please use a separate sheet if necessary)

Python programming, prompt engineering, machine learning modules and basic software engineering.

Additional Knowledge, Skillsets, or Certifications Required

(Please use a separate sheet if necessary)

Multi-Agent System Frameworks: Specific understanding and practical experience with frameworks like A2A (Agent-to-Agent communication), MCP (Model Context Protocol), Langflow, or Flowise, which are central to the project's architecture.

Advanced Data Visualization Libraries/Tools: Proficiency in specific libraries (e.g., Plotly, Matplotlib, D3.js) or dedicated dashboarding tools required for the Dashboard Generation Agent.

Quantitative Analysis and Statistical Methods: Specific expertise in applying various statistical and mathematical methods for data comparison, trend detection, anomaly analysis, and distribution analysis, beyond general machine learning model training.

API Integration and Orchestration: Skills in integrating and orchestrating different APIs and services, crucial for connecting various agents within the multi-agent pipeline.

Domain-Specific Data Understanding: While the project focuses on general data, a deeper understanding of transportation, traffic, car parking, and public transport data structures and nuances at ST Engineering would be beneficial for generating more accurate insights and reports.

Training Required and Provided

(Please use a separate sheet if necessary)

Self-Study and Online Courses: Dedicated self-study through online courses (e.g., Coursera, edX, or specialized platforms) for advanced topics in multi-agent system design, specific data visualization tools, and comprehensive SQL/database management.

Internal Mentorship/Guidance: On-the-job training and mentorship from experienced engineers or data scientists at ST Engineering to gain practical insights into the application of these technologies within the company's data ecosystem, especially concerning their transportation datasets.

Practical Application: Hands-on experience gained through the project's implementation itself, actively applying newly acquired knowledge and skillsets in a real-world context, supported by code reviews and feedback.

D. Project Timeline

(Please use a separate sheet if necessary; a Gantt chart may be included as annex)

Please provide a detailed timeline of the project, including expected output and deliverables

Phase	Task	Duration	Start	End	Dependencies
Planning	Finalize scope, architecture, datasets	Done	Week 1	Week 3	–
Training & Setup	MCP, Stats methods, Agent comms	Done/Ongoing	Week 2	Week 6	–
Agent Development	Quantitative Analysis Agent	2 weeks	Week 6	Week 8	Research & Development
	Insight Agent / Capabilities	2 weeks	Week 8	Week 10	Analysis Agent
	Report Generation Agent / Capabilities	2 weeks	Week 10	Week 12	Insight Agent / Capabilities
	Dashboard Generation Agent / Capabilities	2 weeks	Week 12	Week 14	Report Agent / Capabilities
	Simplified Query & SQL Agents / Capabilities	2 weeks	Week 14	Week 16	Core agents / Capabilities
Integration	Full pipeline integration	2 weeks	Week 16	Week 18	All agents
Evaluation	Agent & system testing	2 weeks	Week 18	Week 20	Integration done
Refinement	Bug fixes & polish	2 weeks	Week 20	Week 22	Evaluation results
Finalization	Demo, documentation, submission	1 week	Week 22	Week 23	Final system ready

E. Supervisor Remarks

(Additional details relevant to the project; please use a separate sheet if necessary)

E.1 Industry Supervisor Remarks

Please provide any other details that are relevant to the project, e.g., references or research papers, description on the complexity of the project, expected learning opportunities, etc.

E.2 Academic Supervisor Remarks

Please provide any other details that are relevant to the project, e.g., references or research papers, description on the complexity of the project, expected learning opportunities, etc.

F. Declaration of Conformity

F.1 Declaration by Industry Supervisor

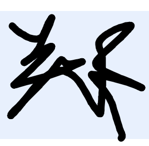
I hereby acknowledge that I have engaged and discussed with the Candidate on the Capstone Project details above, and we are in agreement on the contents provided in the sections above.

Signature

Name of Industry Supervisor: Error! Reference source not found.
Designation: *The industry supervisor's designation*
Department: *The industry supervisor's department / division*
Contact No: Error! Reference source not found.
Email: Error! Reference source not found.
Date: *Click or tap to enter a date*

F.2 Declaration by Candidate

I hereby acknowledge that I have engaged and discussed with my Industry Supervisor on the Capstone Project details above, and we are in agreement on the contents provided in the sections above.



Signature

Name of Candidate: Tay Siang Long Error! Reference source not found.
Contact No: 81618762
Email: 2203190@sit.singaporetech.edu.sg
Date: 11 June 2025 *Click or tap to enter a date*

F.3 Declaration by Academic Supervisor

I hereby acknowledge that I have read and understood the Capstone Project details above. I believe that the proposed project is appropriate to fulfil the requirements for the **ICT4011 Capstone Project** module.

Signature

Name of Academic Supervisor: Error! Reference source not found.
Contact No: *The academic supervisor's contact number*
Email: Error! Reference source not found.
Date: *Click or tap to enter a date*

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